

PRATYAKSHA GUPTA

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EDUCATION

5 year M.Tech in Computer Science and Engineering (specialization in Computational and Data Science)

2022 – 2027
CGPA: 8.24/10

Vellore Institute of Technology

TECHNICAL SKILLS

- Programming & Analytics: Python, SQL, R, Java, Pandas, NumPy
- Databases, BI & Visualization: MySQL, MongoDB, Power BI, Tableau, Excel, Plotly, Matplotlib, Streamlit
- Machine Learning & Frameworks: Scikit-learn, K-Means, Classification, Regression, Predictive Analytics, RAG, FAISS, NLP, Star Schema Modeling

PROJECTS

Airbnb Market Intelligence Dashboard

2026

Tech Stack: Python, Pandas, NumPy, Streamlit, Plotly

- Analyzed 100K+ Airbnb listings using Python, Pandas, and NumPy to uncover pricing trends, demand patterns, and location-level market opportunities across multiple properties.
- Designed 10+ interactive dashboards using Streamlit and Plotly to visualize pricing distribution, occupancy trends, and market concentration, reducing manual reporting effort by 60%.
- Cleaned and transformed 100K+ records to improve data quality and analytical reliability, generating pricing and location-based insights to support business and investment decisions.

Customer Behavior Analytics and Segmentation System

2026

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Processed and analyzed 500K+ retail transactions to identify purchasing patterns, repeat-buying behavior, and revenue-driving trends across a large customer transaction dataset.
- Engineered RFM-based features and segmented 4,000+ customers using K-Means clustering, enabling targeted marketing strategies for different customer groups based on behavior.
- Identified high-value customer segments contributing over 60% of total revenue and improved customer targeting through data-driven segmentation and focused campaign planning.

DocuMind – AI Document Intelligence System

2026

Tech Stack: Python, Streamlit, FAISS, NLP, RAG

- Built a RAG-based document intelligence system to process and query 500+ PDF documents for semantic search, contextual retrieval, and question answering over unstructured content.
- Optimized document retrieval using FAISS vector indexing, reducing search time by approximately 70% and improving the speed and relevance of context-aware information access.
- Developed a Streamlit interface for real-time querying and automated knowledge extraction, converting unstructured documents into a searchable knowledge base for faster information lookup.

CERTIFICATIONS

- Generative AI Professional – Oracle 2025
- AI Foundations – Oracle 2025
- Applied Machine Learning in Python – Coursera 2026